

ARTIFICIAL INTELLIGENCE – ASSESSMENT TOOL 3

U5-A3 REVERSE ENGINEERING

PROJECT TITLE: CHURN GUARD AI

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Course Code : CSA1704

Project Overview

Churn Guard AI is an AI-based customer churn prediction and prevention system designed to identify customers who are likely to stop using a service. The system analyzes customer profile information, usage behavior, transaction history, service interactions, and other relevant features to estimate churn risk. Based on the predicted risk, the system can recommend suitable retention actions such as personalized offers, support follow-ups, or engagement campaigns. For this reverse-engineering assessment, Churn Guard AI is treated as the existing intelligent system to be analyzed. The reverse-engineered design identifies its major modules, data flow, machine-learning pipeline, decision logic, intelligent-agent components, and possible improvements. The assignment questions require system decomposition, architecture reconstruction, algorithm identification, data-flow mapping, and redesign strategies.

1. System Decomposition & Analysis

Churn Guard AI can be decomposed into a set of interconnected modules that transform raw customer information into churn-risk predictions and retention recommendations. The objective is not only to predict whether a customer may churn but also to support proactive intervention.

Major Components

- Customer Data Collection Module – collects customer demographic, account, subscription, usage, billing, transaction, and service-interaction information.

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- Data Preprocessing Module – handles missing values, duplicate records, inconsistent formats, categorical encoding, numerical scaling, and removal of irrelevant attributes.
- Feature Engineering Module – creates useful indicators such as usage frequency, tenure, average spending, complaint frequency, payment behavior, recent activity, and changes in customer engagement.
- Churn Prediction Module – uses a supervised machine-learning model to estimate the probability that a customer will churn.
- Risk Classification Module – converts the predicted probability into categories such as Low, Medium, and High churn risk.
- Intelligent Retention Agent – studies the risk level and customer context and recommends an appropriate retention action.
- Feedback Module – records the outcome of interventions so that future predictions and strategies can be evaluated and improved.
- Dashboard and Notification Module – presents customer risk scores, important contributing factors, and recommended actions to authorized business users.

Underlying AI Logic

- Historical customer records are divided into input features and a target variable representing churn or non-churn.
- The preprocessing pipeline converts raw data into model-ready features.
- A classification model learns the relationship between customer behavior and historical churn.
- The model outputs a churn probability rather than only a yes/no result.
- A business threshold is applied to identify high-risk customers.
- The retention component uses the risk score and customer context to recommend a suitable intervention.
- Actual customer outcomes are later compared with predictions to evaluate precision, recall, accuracy, and business impact.

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Key Constraints

- Customer data may contain missing, noisy, or imbalanced observations.
- False positives can waste retention resources, while false negatives may cause valuable customers to be lost.

The system should provide understandable reasons for high-risk predictions.

- Customer information must be protected through authentication, authorization, and secure data handling.
- The architecture should support increasing numbers of customers and frequent prediction requests.

2. Architecture Reconstruction

The reconstructed architecture of Churn Guard AI follows a layered design in which customer data moves through collection, preprocessing, prediction, risk analysis, intelligent-agent decision-making, and presentation. The modules communicate through structured datasets or APIs.

Reconstructed Architecture

Customer Data	Data Ingestion	Preprocessing & Features	ML Churn Model	Risk Engine	Retention Agent	Dashboard / Action
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Data flow: Customer Data → Data Ingestion → Preprocessing & Feature Engineering → Churn Prediction Model → Risk Classification → Retention Agent → Dashboard/Retention Action → Feedback → Model and Strategy Improvement.

Module Interaction

- The data-ingestion module receives customer records from application databases, CRM systems, transaction systems, or uploaded datasets.
- The preprocessing module validates and transforms the incoming records before they reach the prediction model.

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- The feature-engineering module creates behavior-based indicators that represent customer engagement and risk.
- The machine-learning module receives the feature vector and returns a churn probability. The risk engine maps the probability to a risk level and identifies important contributing factors.
- The retention agent uses the risk information to choose a suitable intervention.
- The dashboard displays the prediction and recommendation to business users.
- The feedback module records whether the customer responded positively, remained active, or churned.

Intelligent Agent Interaction

- Monitoring Agent: watches new customer events and detects significant changes in usage or engagement.
- Prediction Agent: requests or performs churn-risk prediction for relevant customers.
- Retention Agent: selects an intervention based on risk, customer profile, and business rules.
- Feedback Agent: monitors the result of the intervention and sends outcome information for evaluation.
- Coordination Agent: combines information from other agents and prioritizes customers requiring immediate attention.

3. Algorithm Identification

Reverse engineering Churn Guard AI involves identifying the likely machine-learning and data-processing stages used to transform customer records into churn predictions. The exact algorithm cannot be established from the project title alone; therefore, the following is the proposed algorithmic design for Churn Guard AI.

Data Processing Pipeline

- Data Cleaning: remove duplicate records, correct invalid values, and handle missing data.
- Encoding: convert categorical attributes such as contract type, payment method, or service category into numerical representations.
- Feature Scaling: normalize numerical variables when required by the selected model.

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- Feature Selection: retain attributes that provide meaningful information about churn.
- Class Balancing: address cases where churned customers are much fewer than non-churned customers using suitable sampling or class-weight techniques.
- Train-Test Split: separate historical records into training and testing data.
Model Training: train a binary classification model using historical churn labels.
- Prediction: generate a probability of churn for each customer.
- Evaluation: measure performance using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.

Proposed Machine-Learning Algorithm

A suitable primary algorithm for Churn Guard AI is Random Forest classification because churn prediction generally contains a mixture of numerical and categorical-derived features and may involve non-linear relationships. A Random Forest combines multiple decision trees and can provide feature-importance information that is useful for explaining risk factors.

- Input: customer feature vector X .
- Output: churn probability $P(\text{churn} | X)$.
- The model builds multiple decision trees using different samples and feature subsets.
- Each tree produces a classification result and the forest combines the results.
- The final probability is compared with a business-defined threshold.
- For example, a higher probability can place the customer in the High-Risk group, while lower probabilities can be classified as Medium- or Low-Risk.

Decision-Making Logic

- If churn probability is high and the customer shows declining engagement, the system prioritizes immediate retention.
- If risk is medium, the system can recommend personalized engagement or monitoring.
- If risk is low, the system can continue normal engagement without using high-cost retention actions.
- The intelligent agent can combine model risk with business rules, customer value, recent behavior, and previous intervention outcomes.

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4. Data Flow & Process Mapping

The Churn Guard AI data flow begins with customer activity and ends with a risk-aware retention decision. The processing pipeline can operate in batch mode for periodic analysis or near-real-time mode when new customer events are available.

Data Flow

Customer Activity	Data Collection	Distributed Storage	PySpark RDD Processing	Feature Engineering	ML Prediction	Agent Decision	Retention / Feedback
Usage, billing, service, profile	API / DB / events	Historical + current data	map, filter, reduceByKey, join	Behavioral features	Churn probability	Risk + action selection	Action outcome for learning

PySpark RDD Operations

- `map()` transforms each customer record into a structured representation containing the required features.
- `filter()` removes invalid records or selects customers according to analysis criteria.
- `flatMap()` can expand event-level data such as multiple service interactions associated with one customer.
- `reduceByKey()` aggregates customer-level statistics such as number of complaints, total usage, or transaction totals.
- `join()` combines customer behavior with profile, subscription, or historical churn information.
- `groupByKey()` can organize events by customer when grouping is required, although aggregation functions are preferred when possible for efficiency.
- `sortByKey()` can rank customers according to predicted churn risk or priority.

Communication Between Components

- Data ingestion communicates structured customer events to the processing layer.
- The distributed processing layer produces feature records for the prediction service.
- The prediction service returns churn probability and relevant model outputs to the risk engine.
- The risk engine communicates priority information to the retention agent.
- The retention agent communicates recommended actions to the dashboard or business application.
- Feedback from retention actions is returned to the analytics layer for evaluation and future model improvement.

5. Improvement & Redesign

The reverse-engineered Churn Guard AI can be improved by introducing stronger distributed processing, real-time event handling, explainable AI, agent coordination, model monitoring, and optimized resource usage. These improvements can make the system more scalable, efficient, and practical for industry deployment.

Proposed Improvements

- **Distributed Processing:** use PySpark RDDs for large-scale customer-event processing so that records can be partitioned and analyzed in parallel.
- **Real-Time Churn Monitoring:** process recent customer events so that sudden drops in activity or repeated complaints can trigger an updated risk assessment.
- **Feature Store:** maintain reusable customer features so repeated prediction requests do not require complete recomputation.
- **Explainable AI:** show important factors contributing to a high churn score, such as reduced usage, increased complaints, or payment-related changes.
- **Agent-Based Retention:** allow specialized agents to monitor customers, predict risk, recommend actions, and evaluate intervention outcomes.
- **Model Monitoring:** track prediction quality, data drift, class distribution, and changes in customer behavior.
- **Threshold Optimization:** select risk thresholds using business costs so that retention resources are focused on customers where intervention is most valuable.
- **Scalable Storage:** separate historical customer data from frequently accessed operational data and partition large datasets by suitable keys.
- **Fault Tolerance:** use distributed processing and checkpointing/recomputation mechanisms to reduce the impact of node or task failures.
- **Security and Privacy:** apply role-based access, encryption, secure APIs, and controlled access to customer information.

- Continuous Learning: periodically retrain the model using new labeled churn outcomes and intervention feedback.

Redesigned Architecture

Sources	Streaming / Batch	Distributed Data	PySpark RDD	Feature Store	Churn ML + Explainability	Multi-Agent Layer	Action + Feedback
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The redesigned architecture creates a feedback loop: customer behavior → distributed analytics → churn prediction → explanation → intelligent retention action → observed outcome → model/strategy improvement. This allows Churn Guard AI to move from a simple prediction system toward a proactive churn-prevention platform.

Expected Benefits

- Higher scalability for large customer datasets.
- Faster processing through distributed computation and partitioning.
- Earlier identification of customers showing signs of churn.
- More explainable and actionable predictions.
- Better coordination between prediction and retention activities.
- Continuous improvement using actual intervention outcomes.
- Improved operational efficiency by prioritizing high-value, high-risk customers.
- Greater reliability through fault-tolerant distributed processing.

Conclusion

Reverse engineering Churn Guard AI shows how an intelligent churn-management system can be decomposed into data collection, preprocessing, feature engineering, machine-learning prediction, risk classification, intelligent-agent decision-making, and feedback modules. The reconstructed architecture demonstrates a clear flow from customer behavior to churn prediction and retention action. PySpark RDD operations can provide distributed processing for large-scale customer data, while intelligent agents can continuously monitor behavior, coordinate decisions, and adapt retention strategies. The proposed redesign further improves scalability, efficiency, explainability, reliability, and continuous learning, making Churn Guard AI a practical distributed AI solution for customer retention.